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Letter

Self-assembly of zinc 3-hydroxymethyl chlorophyll derivatives esterified with different linear alcohols in deep eutectic solvents

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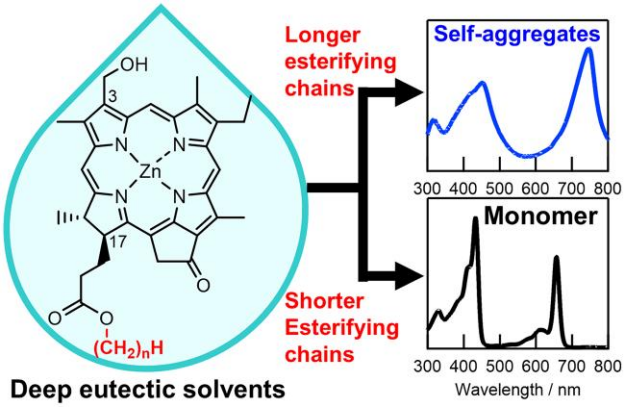
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Abstract

Self-aggregation of Zn 3-hydroxymethyl chlorophyll (Chl) derivatives esterified with different linear alcohols at the 17-propionate residue was investigated in choline chloride-based deep eutectic solvents (DESs). The derivatives esterified with longer alcohols were self-assembled like bacteriochlorophylls in chlorosomes of green photosynthetic bacteria. The physicochemical properties of DESs affected the self-aggregation behaviors of Zn 3-hydroxymethyl Chl derivatives.

Keywords: chlorophyll, chlorosome, deep eutectic solvent.

Graphical Abstract



Photon capture and excitation energy transfer are important for efficient photoenergy conversion in photosynthesis. Green photosynthetic bacteria have a supramolecular light-harvesting complex called chlorosomes for such photoenergy acquisition.[1] In chlorosomes, bacteriochlorophyll (BChl) *c*, *d*, or *e* pigments (Fig. 1A) are self-assembled based on coordination of the 3^1-hydroxy group of one BChl to the central magnesium of another BChl and hydrogen bonding of the 3^1-hydroxy group of one BChl with the 13-keto group of a third BChl.[2] Hydrophobic interactions among esterifying chains at the 17-propionate residue of BChls also contribute to the self-assembly in chlorosomes.[3]

Supramolecular light-harvesting systems based on chlorosome-like pigment self-aggregates have attracted considerable interest because of their importance in not only the elucidation of the photofunctional mechanisms of chlorosomes, but also developments of photofunctional nanosystems for artificial photoenergy conversion.[4] From these viewpoints, in vitro self-aggregates of natural BChls[5–8] and their synthetic model pigments[9–15] have been extensively investigated. These studies have majorly focused on the effects of the pigment molecular structures on the supramolecular structures and spectroscopic properties of the self-aggregates. In contrast, environmental effects on the in vitro self-assembly of natural BChls and their synthetic analogs have attracted less attention; these pigments have been assembled under limited conditions, mainly in hydrophobic environments such as non-polar organic solvents,[5–7,9,10,12,14,15] surfactant micelles,[8,11,13] and lipid layers.[6,11] Although such hydrophobic environments are suitable for the basic pigment interactions (hydrogen bonding

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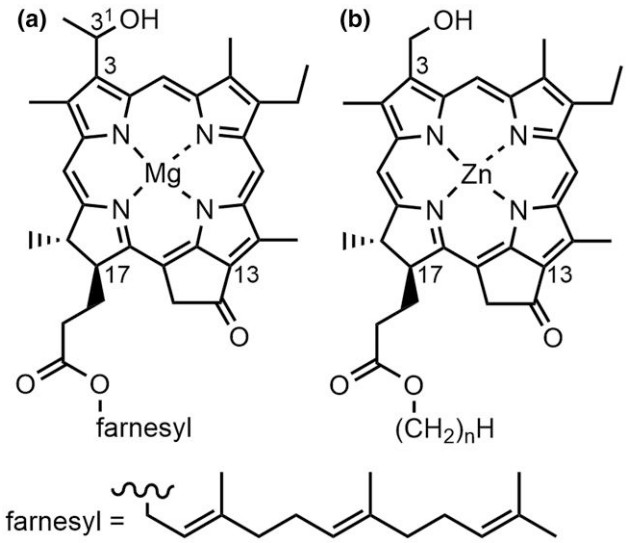


Fig. 1. Molecular structures of BChl *d* in a) green photosynthetic bacteria and b) Zn 3-hydroxymethyl Chl derivatives **1**–**4** (**1**: $n = 1$, **2**: $n = 3$, **3**: $n = 6$, and **4**: $n = 12$) used in this study.

and coordination bonding) in chlorosome-like self-assemblies, other environments for pigment self-aggregates will help in better understanding the architectural principles of these systems and their applications.[16]

Deep eutectic solvents (DESs) are unique liquids that consist of pairs of hydrogen-bond donating and accepting compounds at appropriate molar ratios.[17] DESs are characterized by significant decreases in melting points compared with the neat constituent compounds, and their physicochemical properties can be controlled by the kinds and ratios of the constituent compounds. In this study, we investigated the self-aggregation behaviors of Zn 3-hydroxymethyl chlorophyll (Chl) derivatives **1**–**4** with different esterifying alcohols at the 17-propionate residue (Fig. 1B) in three DESs containing choline chloride as a hydrogen-bond acceptor.

Three DESs were prepared by mixing choline chloride with urea, glycerol, and ethylene glycol at a molar ratio of 1:2 at ca. 35 °C. The three DESs are hereafter denoted as reline, glyceline, and ethaline, respectively.[17] Then, 30 μL of a methanol solution (1 mM) of Zn 3-hydroxymethyl Chl derivatives **1**–**4**, which were synthesized from Chl *a* according to previous reports,[9,13,18,19] were rigorously mixed with 5 g of the DESs and incubated at room temperature for 30 min in the dark, followed by spectral measurements.

Figure 2 shows the electronic absorption spectra of **1**–**4** in reline. Zn 3-hydroxymethyl Chl derivative **1** showed Q_y and Soret absorption bands at 657 and 432 nm, respectively (Fig. 2A), indicating that **1** was monomeric in reline. In the absorption spectrum of **2** in reline, a monomeric Q_y band was also positioned at 657 nm but a small Q_y band was newly observed at 734 nm (Fig. 2B). The red-shifted Q_y band at 734 nm was ascribable to the chlorosome-like self-aggregation of **2**.[5–15] The Q_y band of **3** due to the self-aggregation at 742 nm was larger than that of **2** at 734 nm (Fig. 2C). In the Soret region of the absorption spectrum, a broad band of the self-aggregate of **3** appeared in the longer-wavelength side of the monomeric Soret band. Therefore, the self-aggregate of **3** was co-present with the monomer in reline, and thus **3** is more capable of self-aggregation than **2**. Zn Chl derivatives **4** exhibited the Q_y band at 746 nm with disappearance of the monomeric Q_y band in

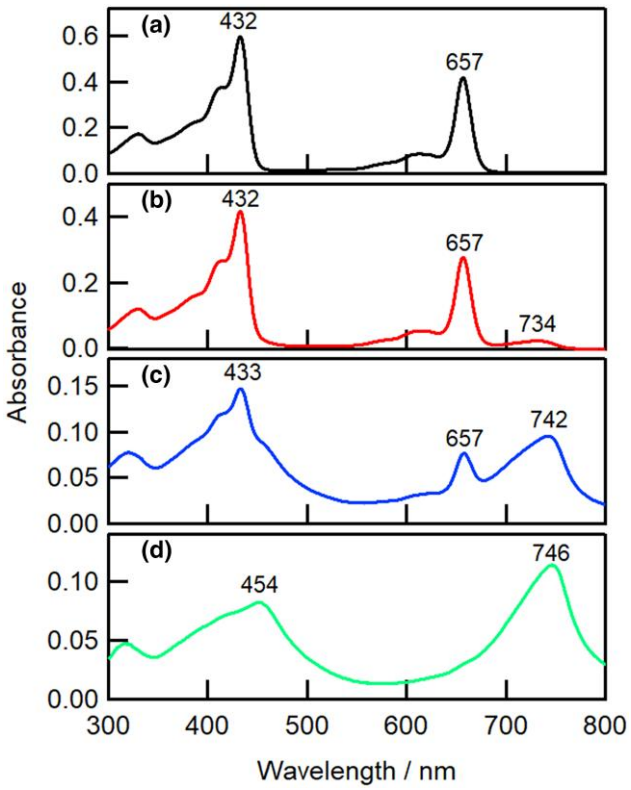


Fig. 2. Electronic absorption spectra of Zn 3-hydroxymethyl Chl derivatives **1**–**4** in reline. The concentrations of **1**–**4** were ca. 7 μM.

reline (Fig. 2D). In addition, the Soret band of **4** was positioned at 454 nm, which was red-shifted relative to the monomeric Soret band. These spectral features are characteristic of a chlorosome-like self-aggregate. These results indicate that elongation of the esterifying alcohol at the 17-propionate residue of Zn 3-hydroxymethyl Chl derivatives promotes chlorosome-like self-aggregation in reline.

In glyceline, **1** and **2** exhibited monomeric Q_y and Soret bands at 657 and 432 nm, respectively (Fig. 3A and B). In the absorption spectrum of **2**, a Q_y band of its self-aggregate was barely observed in the wavelength region between 730 and 740 nm. This spectral feature differs from the presence of the small Q_y band of the self-aggregate of **2** in reline. In contrast, **3** showed two Q_y bands of the monomeric and aggregation forms at 657 and 738 nm, respectively, in glyceline (Fig. 3C). The Q_y band of the self-aggregate of **3** in glyceline was slightly blue-shifted compared with that in reline. The relative absorbance of the Q_y bands of the self-aggregate of **3** in glyceline is smaller that of **3** in reline. The differences in the spectral features of **2** and **3** between those in glyceline and in reline indicate that reline provides a more favorable environment for self-aggregation of the Chl derivatives than glyceline. **4** was self-aggregated in glyceline, judging from the red-shifted Q_y and Soret bands at 743 and 451 nm, respectively (Fig. 3D).

Ethaline gave different spectral features of **1**–**4** from reline and glyceline. The three derivatives **1**–**3** exhibited monomeric absorption spectra, in which the Q_y and Soret bands were positioned at 656 and 432 nm, respectively (Fig. 4A–C). In the absorption spectrum of **4**, two Q_y bands, which were ascribable to the monomeric and aggregation forms, were observed at 656 and 732 nm, respectively (Fig. 4D). The Q_y

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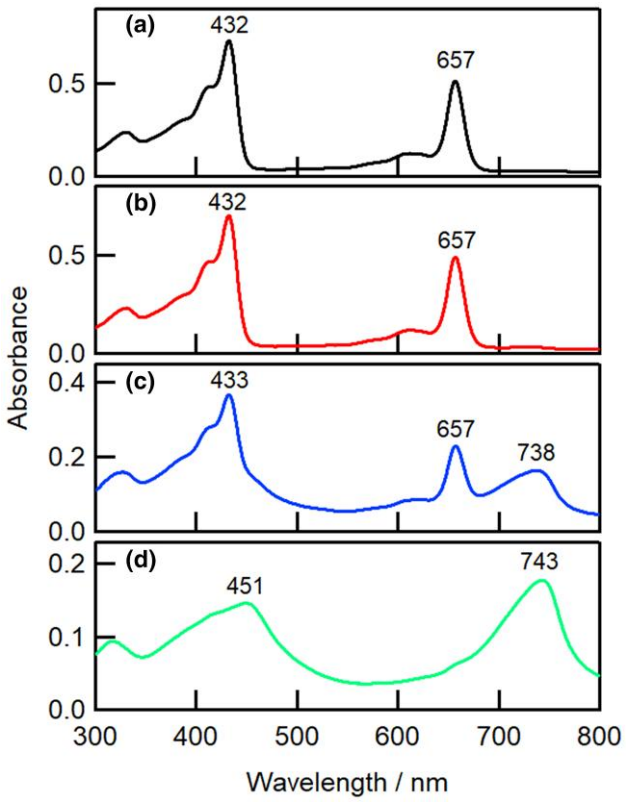


Fig. 3. Electronic absorption spectra of Zn 3-hydroxymethyl Chl derivatives **1**–**4** in glyceline. The concentrations of **1**–**4** were ca. 7 μM.

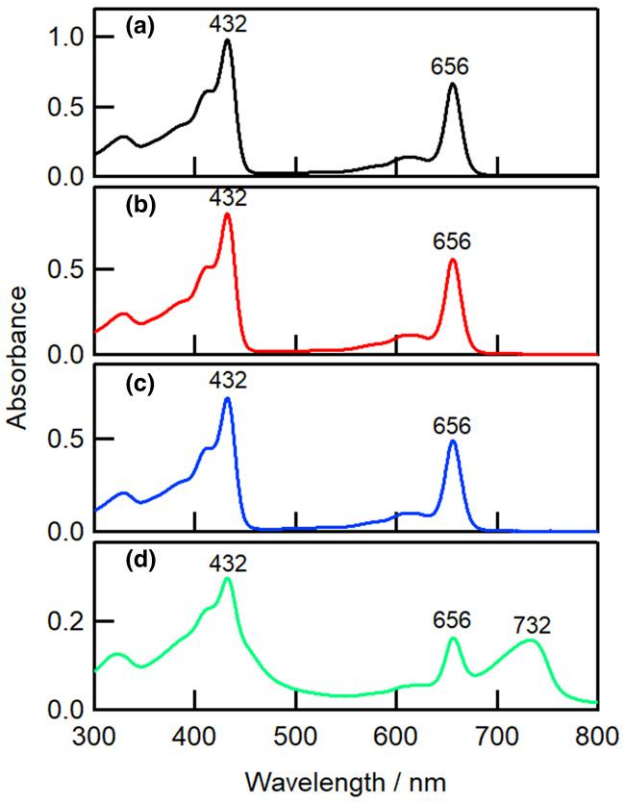


Fig. 4. Electronic absorption spectra of Zn 3-hydroxymethyl Chl derivatives **1**–**4** in ethaline. The concentrations of **1**–**4** were ca. 7 μM.

position of the self-aggregate of **4** in ethaline was shifted to a shorter wavelength than that of **4** in reline and glyceline. These results indicate that ethaline is less favorable for chlorosome-like self-aggregation of Zn 3-hydroxymethyl Chl derivatives than reline and glyceline.

Circular dichroism (CD) spectroscopy supported the results of the aggregation behaviors of **1**–**4** in the three DESs demonstrated by absorption spectroscopy. Zn 3-hydroxymethyl Chl derivatives **1** and **2** exhibited small negative CD signals at 656 nm in reline (Supplementary Figs. S1A and B). These CD signals correspond to the monomeric Q_y band in the absorption spectra. In contrast, the self-aggregates of **3** and **4** in reline showed an *S*-shaped CD signal with a positive signal at its longer-wavelength side between 700 and 800 nm (Supplementary Figs. S1C and D). The CD signatures of **1**–**4** in glyceline were similar to those in reline; **1** and **2** had small negative CD signals due to the monomeric form whereas **3** and **4** showed a combination of an *S*-shaped CD signal and a positive signal (Supplementary Fig. S2). The *S*-shaped CD signal arises from chlorosome-like self-aggregates of **3** and **4**, as such a signal is characteristic of BChl self-aggregates in chlorosomes.[20] The combination of the *S*-shaped CD signal with a positive signal observed here would have originated from an overlap of two *S*-shaped signals. These results suggest that **3** and **4** formed at least two aggregation species in the reline and glyceline prepared in this study. In ethaline, **1**–**3** exhibited negative CD signals at 656–657 nm (Supplementary Fig. S3A–C), indicating that these derivatives are present as a monomer. After a while, **4** in ethaline showed an *S*-shaped CD signal between 700 and 800 nm due to its chlorosome-like self-aggregates (Supplementary Fig. S3D). The current spectroscopic analyses indicate that the self-aggregation of Zn 3-hydroxymethyl derivatives in the three DESs is promoted in the order of reline > glyceline > ethaline.

The results in this study indicate the importance of two factors for the self-aggregation of Zn 3-hydroxymethyl Chl derivatives in DESs; one is the physicochemical properties of DESs and the other is the esterifying chains of the derivatives. Generally, hydrophobic environments are advantageous for the chlorosome-like self-aggregation of natural BChls *c*, *d*, and *e*, and their model pigments,[5–15] whereas these pigments tend to be monomeric in polar organic solvents. Indeed, **1**–**4** showed absorption spectra that were characteristic of the monomer in methanol (Supplementary Fig. S4). DESs are rather polar solvents, judging from the solvatochromic spectroscopic responses[21,22]; the normalized polarity parameters E_TN, which range between 0 (non-polar) and 1 (highly polar),[23] of reline, glyceline, and ethaline are estimated to be 0.81, 0.84, and 0.83.[22] The E_TN value of reline that is smaller than that of glyceline and ethaline is in line with the self-aggregation behaviors of Zn Chl derivatives in reline compared with those in glyceline and ethaline. In contrast, the difference between the self-aggregation behaviors in glyceline and ethaline cannot be simply explained by the solvent polarities, as the E_TN value of glyceline is larger than that of ethaline. Kamlet–Taft parameters[24] of DESs would be helpful when discussing the self-aggregation behaviors of Zn Chl derivatives in DESs. Kamlet–Taft parameters include the parameters of hydrogen-bonding donor ability (α), hydrogen-bonding acceptor ability (β), and dipolarity/polarizability (π*) (Supplementary Table S1). Among the three parameters, the parameters concerning the hydrogen-bonding formation,

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α and β, would affect the formation of the self-aggregates and the β values (0.50, 0.52, and 0.57 for reline, glyceline, and ethaline, respectively[22]) match the order of the promotion of the self-aggregation in the three DESs (reline > glyceline > ethaline). Therefore, a decrease in the hydrogen-bonding acceptor ability of DESs would be important for the self-aggregation of Zn Chl derivatives in DESs. This assumption can be rationalized by the decrease in the inhibition effects of the key interactions (hydrogen bonding and coordination bonding) between Zn Chl derivatives in chlorosome-like self-assembly caused by the decrease in the hydrogen-bonding acceptor ability of the DESs. In contrast, the α values are not consistent with the order of the promotion of the self-aggregation. This might be due to a limited inhibition effect on the hydrogen-bonding formation between the 3^1-hydroxy and 13-keto groups between the Zn Chl derivatives. The π^* parameter would affect interactions concerning dipoles in the self-aggregation and the effects of such interactions might be smaller than those of hydrogen bonding and coordination bonding for the self-aggregation.

The viscosity of DESs would be another important property to influence the self-aggregation of Zn Chl derivatives, although there is little information on the effects of the viscosity of liquids on chlorosome-like self-assembly. The order of the viscosities of the three DESs (Supplementary Table S1) is qualitatively consistent with the promotion of the self-aggregation of the derivatives. The tendency of the self-aggregation in higher viscous environments is in line with the comment in a previous report.[25]

The self-aggregation of Zn 3-hydroxymethyl Chl derivatives in the three DESs depended on the length of their esterifying alcohols at the 17-propionate residue; longer esterifying chains were advantageous for the self-aggregation. These results suggest that hydrophobic interaction among the esterifying chains of these derivatives plays a role in the self-aggregation in the DESs. Relationships between the relative ratios of the Q_y absorbance of the self-aggregates of **1**–**4** in the DESs and the carbon numbers of the esterifying alcohols (Supplementary Fig. S5) show roughly similar slopes during the phase in which the relative ratios increase. These results suggest that the roles of the esterifying alcohols of **1**–**4** in the self-aggregation in the DESs is analogous.

In addition to the hydrogen-bonding network, DESs are likely to consist of complicated interplays, with electronic interaction between cationic and anionic species, and the steric effects of the composites and their geometries.[17] Such complexity results in the presence of heterogeneous domain structures in DESs and might produce heterogeneous self-aggregates of Zn 3-hydroxymethyl Chl derivatives in the DESs.

To summarize, this study first demonstrates the self-aggregation behaviors of Zn 3-hydroxymethyl Chl derivatives in DESs and indicates that the self-aggregation is sensitive to the physicochemical properties of the DESs and the esterifying chains of the Chl derivatives. The current results will be useful for understanding the structural and photofunctional mechanisms of pigment self-aggregates in chlorosomes of green photosynthetic bacteria. Additionally, this study will also contribute to the application of DESs for photoenergy conversions.

Supplementary data

Supplementary material is available at *Chemistry Letters* online.

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Conflict of interest statement. None declared.

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